were, stretched or expanded by material substance, and absolutely empty space had been inconceivable. But now, empty space was allotted a very important function: it became the vehicle for geometry and kinematics, by making possible the various arrangements and movements of atoms.

Although the atom was regarded as having a special position in space, also a shape, and as executing certain movements, it was not allotted any attribute other than these purely geometrical properties. The atom had neither colour nor smell nor taste, and the properties perceptible by human senses, together with their changes and mutations, were supposed to be produced by the movement and displacement of atoms in space. Just as both tragedy and comedy could be written with the same latters of the alphabet, the vast variety of events in the universe were regarded as the products of the selfsame atoms, of their different positions and different motions. Democritus said: ‘A thing merely *appears* to have colour; it merely *appears* to be sweet or bitter. Only atoms and empty space have a real existence.’

The basic ideas of atomic theory were taken over and modified, in part, by the later Greek philosophers. Plato, in his dialogue *Timaeus*, co-ordinates these ideas with Pythagoras’ theory of the harmony of numbers, and identifies the atoms of the elements—earth, air, fire and water—with the symmetrical bodies, cube, octahedron, tetrahedron and ikosahedron. The Epicureans, too, adopted the essential concepts of the atomic theory, and appended to it an idea which was to play an important part in natural science at a later date: the idea of natural necessity. According to this theory, the atoms are not thrown together arbitrarily, at random, like dice, nor set in motion by forces such as Love or Hate, but their paths are determined by natural laws, or by the working of blind necessity.

After this point, there was no further development in atomic theory, either in the philosophy or in the science of Antiquity.

II. MODERN ATOMIC THEORY, UP TO THE END OF THE NINETEENTH CENTURY

All the progress which we have mentioned, occurred in the course of a few centuries. Two thousand years elapsed then before they were recalled, and before another thinker took up these ideas and transformed them into something fruitful. During the latter part of Antiquity, and during the Middle Ages in particular, the philosophy of Aristotle was accepted as an incontestable foundation, and for the Christian outlook reality had changed to such an extent that the attention of mankind was not attracted by material Nature for a long time.

The first philosopher to revive these neglected trends of thought was the